Question # 9 of 10 (Start time: 06:20:33 PM, 16 January 2020) Total Marks: 1

Which of the following statements is true about discrete topological space?

Select the correct option

Reload Math Equations

○	Every discrete topological space is not regular.
○	Every discrete topological space is regular.
○	None of these
○	Every discrete topological space may or may not be regular.

Click to Save Answer & Move to Next Question

Which of the following is not an example of compact spaces?

Select the correct option

Reload Math Equations

- A finite set with any toplogy.
- A set with topology containing finite number of elements.
- An infinite set with discrete topology.
- A set with indiscrete toplogy.



Click to Save Answer & Move to Next Question

Which of the following is a correct reason why every subspace of a Hausdorff space is also Hausdorff?

elect the correct option

Reload Math Equations

- Because every subspace of a topological space is always Hausdorff.
- Because the subspace have different separation properties from the parent space.
- Because the subspace inherits the separation properties from the parent space.
- Because every subspace is closed in a Hausdorff space.

Given a Hausdorff space (X, T) and a subset $Y \subset X$, with the subspace topology (Y, T_Y), which of the following statements is true?

Select the correct option

Reload Math Equations

- Y must be connected.
- Y with the subspace topology (Y, T_Y) is also Hausdorff.
- Y must be compact
- Y with the subspace topology (Y, T_Y) is not Hausdorff.

Click to Save Answer & Move to Next Question

A topological space X is said to be separable if it has a countable dense subset. Which of the following is a countable dense subset of the real numbers?

Select the correct option

Reload Math Equations

○	The set of rational numbers
○	The set of integers
○	The set of all real numbers
○	The set of natural numbers

Click to Save Answer & Move to Next Question

Let X be a topological space. Which of the following statement is true?

Select the correct option

Reload Math Equations

- Every pair of disjoint closed sets in X can be separated by open sets.
- Every pair of disjoint closed sets in X can be separated by closed sets.

Click to Save Answer & Move to Next Question

Which of the following statements about an indiscrete topological space is true?

Select the correct option

Reload Math Equations

- It is always a T_0 space.
- It is never a T_0 space.
- None of these
- It may be a T_0 space.

Question # 4 of 10 (Start time: 09:04:45 PM, 10 January 2025)

Which of the following statements is true about Hausdorff spaces?

Select the correct option

- ○ Every pair of distinct points can be separated by disjoint open sets.
- ○ Every open set is dense.
- ○ Every sequence has a convergent subsequence.
- ○ Every pair of distinct points can be separated by overlapping open sets.

Which of the following statements correctly compares T_1 and T_0 properties?

Select the correct option

○	T_0 property implies T_1 property, but not vice versa.
○	T_0 property is a stronger property than T_1 property.
○	T_1 property is equivalent to T_0 property.
○	T_1 property implies T_0 property, but not vice versa. ✓

Which of the following statements is true about discrete topological space?

Select the correct option

- ○ Every discrete topological space may or may not be regular.
- ○ None of these
- ○ Every discrete topological space is regular.
- ○ Every discrete topological space is not regular.

If (X, τ) be a separable topology then it must have countable dense set.

Select the correct option

- ○ True



- ○ False

MTH634 - Topology (Quiz No. 4) Quiz Start Time: 09:21 PM, 10 January 2025

Question # 3 of 10 (Start time: 09:25:38 PM, 10 January 2025) Total Marks: 1

Which of the following space is a Hausdorff?

Select the correct option

Reload Math Equations

- An infinite set with cofinite topology.
- Real numbers with topology generated by $\{(a, \infty) | a \in R\}$
- Real numbers with Lower limit topology.
- A non empty set X with indiscrete space.

Click to Save Answer & Move to Next Question

Which of the following statements is equivalent to the statement that a topological space X is T1?

Select the correct option

Reload Math Equations

○	None
○	For any two distinct points x and y in X, there exist disjoint open sets U and V such that $x \in U$ and $y \in V$.
○	For any two distinct points x and y in X, there exists an open set U such that $x \in U$ and $y \notin U$.
○	For any two distinct points x and y in X, there exists an open set V such that $y \in V$ and $x \notin V$.

Click to Save Answer & Move to Next Question

Which of the following is an example of a metric space that is also Hausdorff?

Select the correct option

- The discrete metric space on any set.
- The set of complex numbers $\mathbb{C}$ with the metric defined on the set of complex numbers.
- All of these
- The set of real numbers $\mathbb{R}$ with the standard Euclidean metric.

Which of the following statements is equivalent to the statement that a topological space X is regular?

Select the correct option

Reload Math Equations

- For any point x in X and any closed set F such that $x \notin F$, there exists a closed set G such that $x \in G$ and $F \cap G = \emptyset$.
- A topological space X is said to be regular if, for every point x in X and every closed set F in X such that $x \notin F$, there exist disjoint open sets U and V in X such that $x \in U$ and $F \subseteq V$.

Click to Save Answer & Move to Next Question

Which of the following topological spaces is Hausdorff?

Select the correct option

Reload Math Equations

- The set of all points in the plane with the Euclidean topology.
- The set of all real numbers with the usual topology.
- The set of all finite sets with the discrete topology.
- The set of all infinite sets with the discrete topology.

Click to Save Answer & Move to Next Question

Consider a set X with the indiscrete topology. What can be said about the singleton sets in X?

Select the correct option

Reload Math Equations

- They are closed but not open.
- They are open but not closed.
- They are neither open nor closed.
- They are open and closed.

Click to Save Answer & Move to Next Question

Which of the following statements is true about Hausdorff spaces?

Select the correct option

Reload Math Equations

- Every open set is dense.
- Every pair of distinct points can be separated by disjoint open sets.
- Every pair of distinct points can be separated by overlapping open sets.
- Every sequence has a convergent subsequence.

Click to Save Answer & Move to Next Question

Let X be a topological space. Which of the following statement is true?

Select the correct option

Reload Math Equations

- Every pair of disjoint closed sets in X can be separated by closed sets.
- Every pair of disjoint closed sets in X can be separated by open sets.



Click to Save Answer & Move to Next Question

A topological space is a T_1 - space if and only if each of its finite subset is ...

Select the correct option

Reload Math Equations

- neither open nor closed.
- an open set
- both open and closed.
- a closed set.

Click to Save Answer & Move to Next Question

Which of the following statements correctly characterizes a T_1 topological space?

Select the correct option

Reload Math Equations

- Every subset is closed.
- Every subset is open.
- Every singleton subset is closed.
- Every singleton subset is open.

Click to Save Answer & Move to Next Question

Which of the following statements is true about metric spaces?

Select the correct option

Reload Math Equations

- Every metric space is discrete.
- Every metric space is Hausdorff.
- Every metric space is not Hausdorff.
- Every metric space is complete.



Click to Save Answer & Move to Next Question